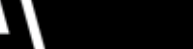




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The logo for St. Augustine State, featuring the word "STATE" in a large, red, blocky font with a white outline. The letter "A" is significantly larger and more stylized than the others, with a white outline and a red fill. The letters "S", "T", "U", "G", "E" are smaller and also in red with white outlines. The logo is set against a black background.

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- Antimicrobial resistance (AMR) is a critical global health threat, causing over 2.8 million infections and 35,000 deaths annually in the U.S.
- Current susceptibility testing evaluates many antibiotics per isolate, making the process costly and time-consuming
- Although genomic data can predict antimicrobial resistance, existing models are not yet reliable enough for clinical decision-making.
- We aim to prioritize the most informative antibiotics using machine learning to improve testing efficiency

- AMR metadata collection by ~BV-BRC (Bacterial and Viral Bioinformatics Center)

- In genome name column, *Escherichia coli*
- In antibiotic column, multiple antibiotics (e.g., ampicillin, ciprofloxacin, cefotaxime, etc.)
- Focus on susceptible samples for model training and analysis

- Feature Design

- 562.48345.fna -> 562.48345.fna.txt
 - Genome sequences are transformed into n-gram based features for machine learning models



gtgtgccgtc
gttgccgtca
ttgccgtcac
tgcgcgtaca
gccgtcacag
ccgtcacagc
cgctcacagg
gtcacagcct
tcacagcctc
cacagcctcg
acagcctcga
cagcctcgat
agcctcgata
cctcgatat
ctcgatataa
tcgatataat
cgatataatc
gatataatcc
atataatcca

- Details of the dataset used for the experiment

Antibiotic	Susceptible
ampicillin	2696
ciprofloxacin	5148
amoxicillin/clavulanic acid	1151
cefotaxime	3503
trimethoprim/sulfamethoxazole	2579
ceftazidime	5750

```

graph LR
    AMR[(AMR Dataset)] --> IF[Intermediate File]
    subgraph Preprocessing
        IF --> STK[Select Top K-mers]
        IF --> GK[Generate K-mers]
        GK --> CFK[Calculate the frequency of K-mers]
        STK --> EFI[Employ Feature Importance]
        CFK --> EDA[Employ Data Augmentation]
    end
    EFI --> ML[ML Models: RF, CNN, MLP, LGBM, XGB]
    EDA --> ML
  
```

- We utilize n-gram frequency analysis.
- The detailed step is below:
 - ❑ Step 1. Download AMR Dataset files from the BV-BRC repository.
 - ❑ Step 2. Select susceptibility data for six antibiotics associated with *Escherichia coli* (1 = susceptible, 0 = otherwise).
 - ❑ Step 3. Make intermediate files by filtering out meta information.
 - ❑ Step 4. Generate k-mers (i.e., 7 or 10-mers) from each file.
 - ❑ Step 5. Calculate the frequency of each k-mers across all files and select the top 1000 features out of 1 million features.
 - ❑ Step 6. Construct a multi-label dataset (CSV) containing susceptibility labels for multiple antibiotics for each genome.
 - ❑ Step 7. Train and evaluate multiple machine learning models (RF, CNN, MLP, LightGBM, XGBoost) using 5-fold cross-validation, and analyze performance by adjusting classification thresholds.
 - ❑ Step 8. Based on the test results, prioritize effective antibiotics for each genome.

● Model Performance & Threshold changing results

- Model performance reaches up to approximately 78% in F1-score across different models.

Model	F1-score	ROC-AUC
Random Forest	0.7628	0.8326
CNN	0.6868	0.7512
MLP	0.7786	0.8384
LightGBM	0.7801	0.8446
XGBoost	0.7812	0.8458

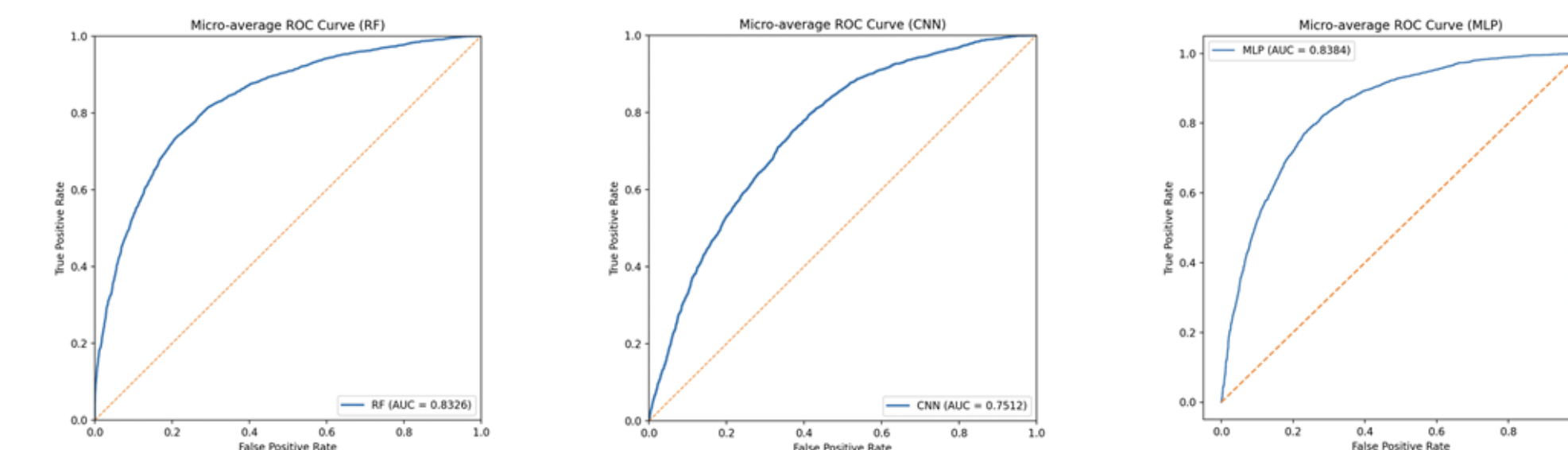
Model	F1-score (Threshold changed)	ROC-AUC
Random Forest	0.7789	0.8326
CNN	0.7654	0.7512
MLP	0.7829	0.8384
LightGBM	0.7639	0.8446
XGBoost	0.7515	0.8458

- **Top-3 antibiotics prediction**
 - The model ranks antibiotics by predicted susceptibility and selects the top-3 candidates for each genome, enabling targeted antibiotic prioritization.

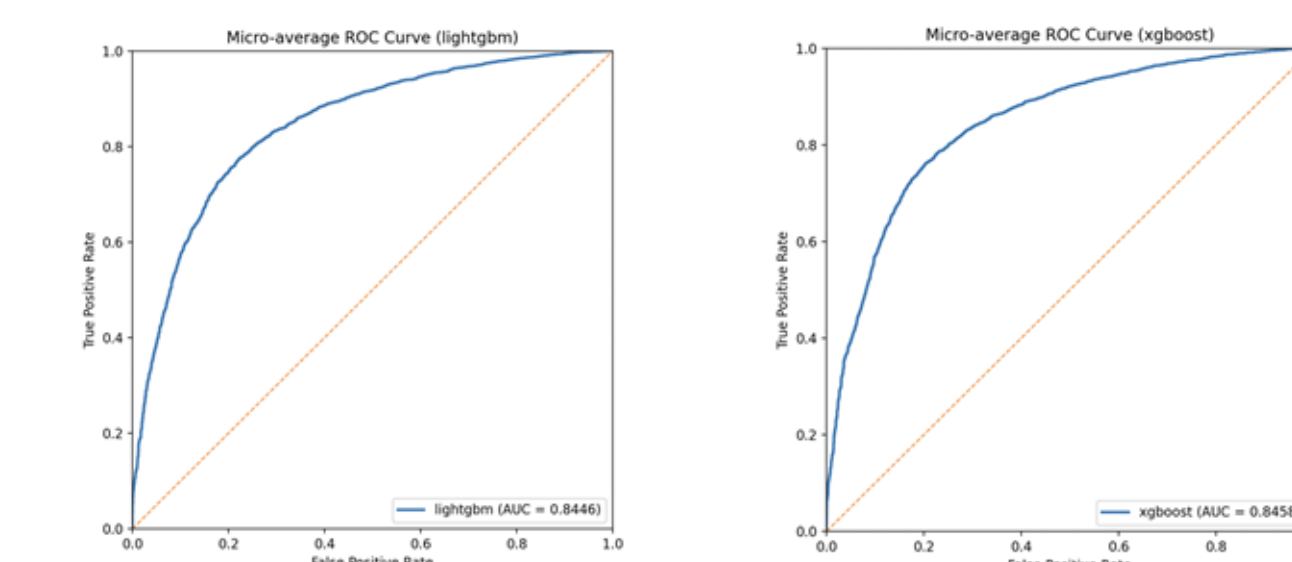
[illegible]

→ The model successfully identifies the most informative antibiotics, enabling targeted susceptibility testing.

- ROC curves indicate that most models achieve strong discriminative performance across genome-antibiotic combinations.

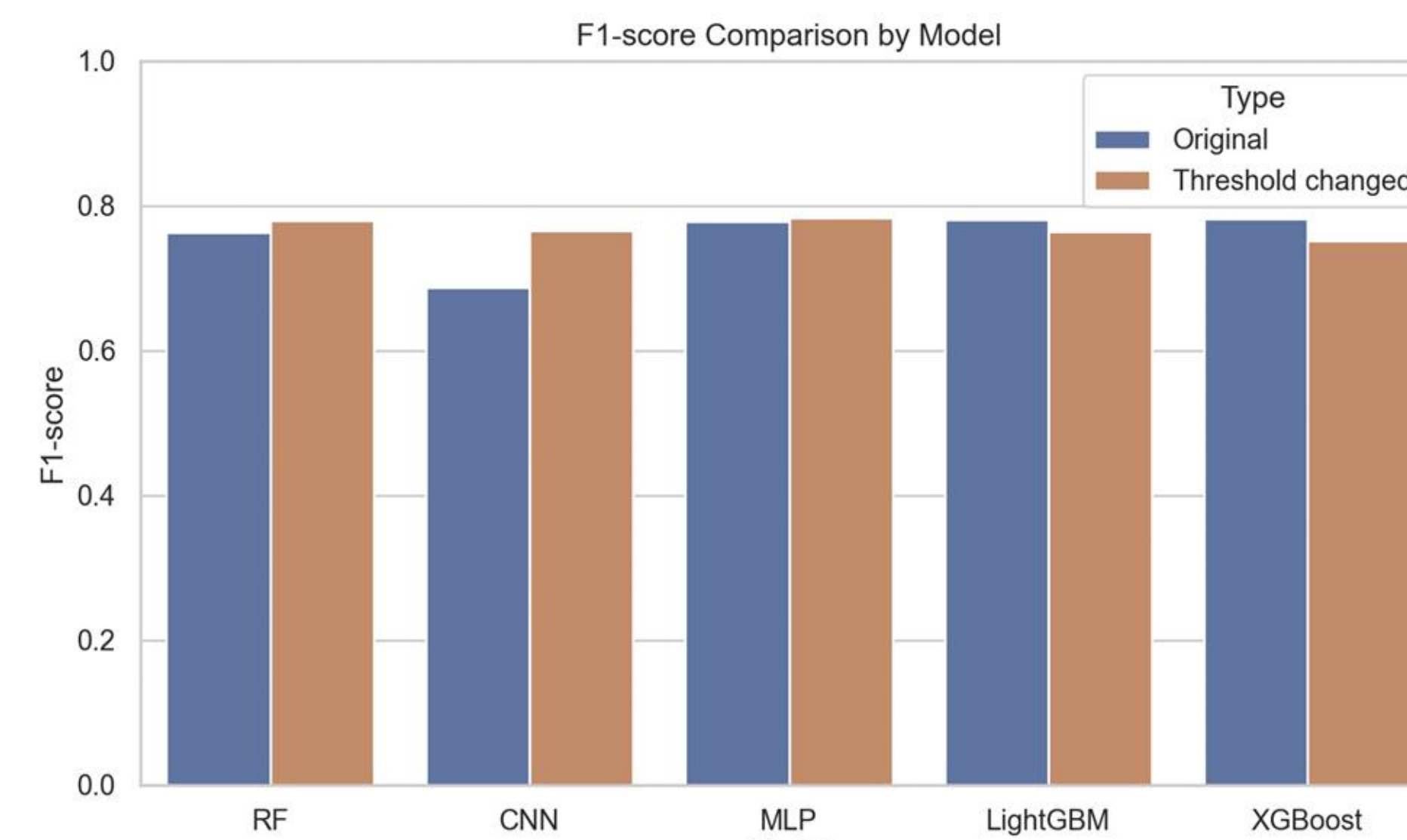


✓ ROC curves show strong discriminative performance with high AUC values across genome-antibiotic combinations.



✓ In most cases, our results match or exceed previously reported performance, indicating robust prediction capability.

- F1-score comparisons show that performance varies across models and is influenced by threshold selection.
 - Since this is a multi-label classification problem, a fixed threshold (e.g., 0.5) may not be optimal for all antibiotics.



- F1-scores reach up to ~0.78 across models, demonstrating stable performance in multi-label prediction.
- Threshold tuning improves performance in neural network-based models.

Conclusion

- In this study, we used n-gram features from whole-genome data to perform multi-label classification and prioritize effective antibiotics for each genome.
- The models achieved stable performance, with F1-scores reaching up to 78%, and showed further improvements with threshold changing.
- These results demonstrate that genomic feature-based approaches can improve the efficiency of antibiotic selection, with potential for future clinical application.



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